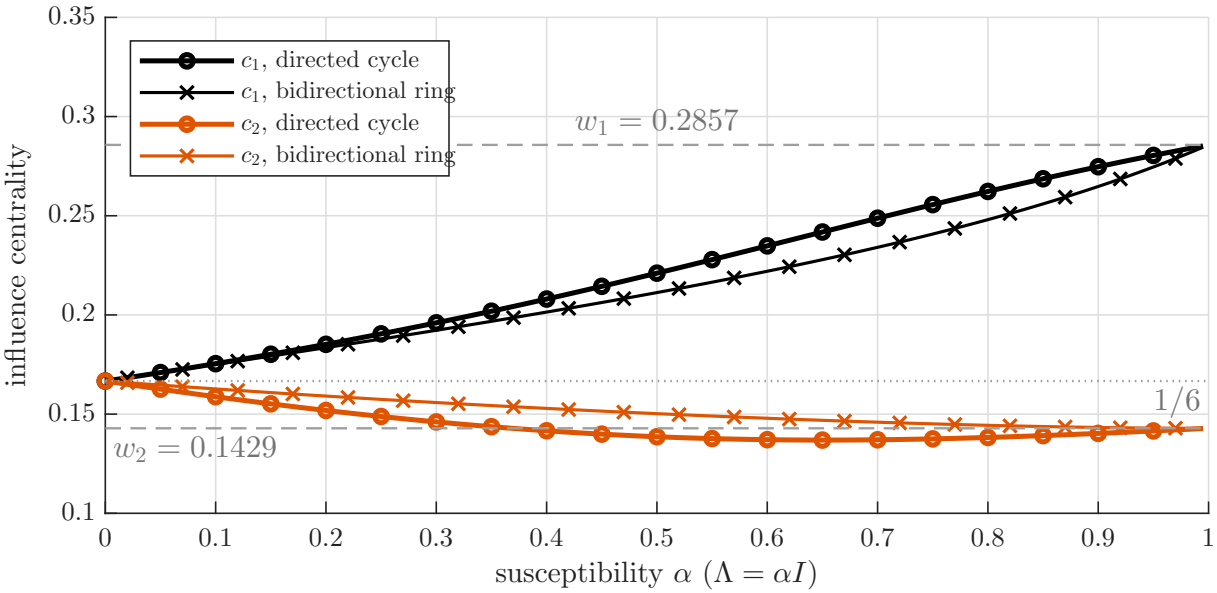


Figure 13: Influence centrality c_1 and c_2 in the FJ model with $W = A(1/2)$ and $\Lambda = \alpha I$, $\alpha \in [0, 0.995]$, and $u = x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$, on the directed cycle (dots) and on the bidirectional ring (crosses). The dashed lines mark the social power w_1 and w_2 of the FDG model with the same matrix $A(1/2)$, the dotted line the uniform value $1/6$.

For $\alpha = 0$ every agent sticks to its own opinion, $V_0 = I$, and the influence is equally shared, $c_0 = \mathbf{1}_n/n$. As $\alpha \to 1$ the agents barely listen to their prejudices and the FJ model behaves as the FDG model: $V_\alpha = (1-\alpha)(I - \alpha W)^{-1} \to \mathbf{1}_n w^\top$ [5], the final opinions approach the FDG consensus, and $c_\alpha \to w$, since in a common opinion the mean weight of $x_i(0)$ and its weight in the consensus coincide. The limit needs W fully regular, here $A(1/2)$, which is primitive (Corollary 1.14); at $\alpha = 1$, $I - W$ is singular and c is not defined, so the two measures meet only as a limit. In Figure 13, c_1 grows from $1/6$ to $w_1 = 2/7$ and c_2 decreases to $w_2 = 1/7$: as the agents open up to the others, the inertia of agent 1 counts more. On the directed cycle c_2 first goes below w_2 (about 0.137 at $\alpha \approx 0.6$).

4 Conclusions

Table 3 collects the comparison between the two models developed in Sections 2 and 3.

	FDG model (1)	FJ model (4)
Resistance to the others	self-weight a_{ii}: inertia, a weight on the current opinion	attachment $1-\lambda_i$ to the prejudice u_i: an anchor to a fixed value
Stubborn agent (Definition 2.4)	$a_{ii}=1$: a condition on the matrix only	$\lambda_i=0$ and $u_i=x_i(0)$: a condition on the matrix and on the input
Opinion of a stubborn agent	a state	an input
Convergence	Table 1: sinks of $C(G)$ and their aperiodicity	all agents P-dependent (Corollary 3.3), or W^{22} regular (Theorem 3.2)
Asymptotic stability	never: the eigenvalue 1 lies on the unit circle (Theorem 1.7)	if and only if all agents are P-dependent, i.e. $\rho(\Lambda W)<1$ (Corollary 3.3)
Final opinions	consensus $(w^\top x(0))\mathbf{1}_n$ (Theorem 2.1), or disagreement with at least two aperiodic sinks (Theorem 2.3)	if all agents are P-dependent, $x(\infty)=Vu$ (9): convex combinations of the prejudices, independent of $x(0)$
Strongly connected network	consensus or no convergence, never disagreement	disagreement whenever the prejudices that count differ
Aperiodicity	always needed	needed only for the P-independent agents
Measure of power	social power w (2): requires a consensus	influence centrality c (12): requires $\rho(\Lambda W)<1$ and $u=x(0)$
From one model to the other	with stubborn agents, an FJ model with $\lambda_i=0$, $u_i=x_i(0)$ for the stubborn agents [5]	an FDG model on the augmented graph (11), with n virtual stubborn agents

Table 3: Comparison between the FDG model and the FJ model.

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